

Project Design Phase
Problem – Solution Fit

Date	20 February 2026
Team ID	LTVIP2026TMIDS34731
Project Name	HouseHunt-Finding Your Prefect Rental Home
Maximum Marks	2 Marks

Problem – Solution Fit Template

Problem Statement

Manual and unstructured property searching, owner onboarding, and booking management continue to create inefficiencies in rental housing platforms.

Renters, property owners, and platform administrators face significant challenges due to the absence of a centralized, organized system for property listing, booking tracking, and platform governance.

! Key Problems Identified

- ✗ Renters struggle with scattered property information, poor filtering systems, and lack of real-time availability updates.
 - ✗ Property owners lack a simple, self-service interface to manage listings, update availability, and respond to booking inquiries.
 - ✗ Admins require better monitoring tools to verify owners, manage users, and maintain listing quality.
 - ✗ Many rental platforms are not optimized for small or independent property owners who want to digitize their rental process.
 - ✗ Lack of structured booking workflows leads to confusion in confirmation, negotiation, and lease finalization.
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Proposed Solution: HouseHunt

HouseHunt is a role-based, full-stack MERN web application built using **MongoDB, Express.js, React.js, and Node.js**.

- ✓ A clean, responsive user interface for Renters, Owners, and Admins
- ✓ JWT-based authentication with secure role-based dashboards
- ✓ Advanced property search and filtering system

- ✓ Booking request and approval workflow
 - ✓ Property CRUD (Create, Read, Update, Delete) functionality for owners
 - ✓ Admin panel for user verification and platform governance
 - ✓ Scalable backend architecture for future expansion (payments, reviews, analytics)
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Purpose of the Solution

- ✓ Centralize rental property operations into a structured digital ecosystem
 - ✓ Empower independent property owners with a self-service property management dashboard
 - ✓ Simplify the rental discovery and booking experience for renters
 - ✓ Reduce administrative workload through role-based control and automated workflows
 - ✓ Improve user experience with responsive UI, fast API responses, and clear booking status tracking
 - ✓ Ensure transparency between renter and owner through structured inquiry and confirmation system
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Value Proposition

HouseHunt eliminates inefficiencies in traditional rental processes by:

- Digitizing property listings
- Automating booking approvals
- Providing real-time dashboard updates
- Enforcing admin-level governance
- Ensuring secure role-based access control

The platform bridges the gap between renters and independent property owners through a secure, scalable, and user-friendly rental ecosystem.

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one; explore different sides.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; Indirectly associated: customers spend free time on volunteering work (I.e. Greenpeace)	
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure > confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	